Effects of intravenous and dietary lipid challenge on intramyocellular lipid content and the relation with insulin sensitivity in humans.

An increased intramyocellular lipid (IMCL) content, as quantified by (1)H-magnetic resonance spectroscopy ((1)H-MRS), is associated with reduced insulin sensitivity. At present, it is unclear which factors determine IMCL formation and how rapidly IMCL accumulation can be induced. We therefore studied the impact of hyperinsulinemia and elevated circulating nonesterified fatty acid (NEFA) levels on IMCL formation and insulin sensitivity. We further evaluated the influence of a high-fat diet on IMCL storage. In the infusion protocol, 12 healthy male subjects underwent a 6-h hyperinsulinemic-euglycemic glucose clamp with concomitant infusion of Intralipid plus heparin. IMCL was quantified by (1)H-MRS in soleus (SOL) and tibialis anterior (TA) muscle at baseline and then every hour. IMCL levels started to increase significantly after 2 h, reaching a maximum of 120.8 +/- 3.4% (SOL) and 164.2 +/- 13.8% (TA) of baseline after 6 h (both P < 0.05). In parallel, the glucose infusion rate (GIR) decreased progressively, reaching a minimum of 60.4 +/- 5.4% of baseline after 6 h. Over time, the GIR was strongly correlated with IMCL in TA (r = -0.98, P < or = 0.003) and SOL muscle (r = -0.97, P < or = 0.005). In the diet protocol, 12 male subjects ingested both a high-fat and low-fat diet for 3 days each. Before and after completion of each diet, IMCL levels and insulin sensitivity were assessed. After the high-fat diet, IMCL levels increased significantly in TA muscle (to 148.0 +/- 16.9% of baseline; P = 0.005), but not in SOL muscle (to 114.4 +/- 8.2% of baseline; NS). Insulin sensitivity decreased to 83.3 +/- 5.6% of baseline (P = 0.033). There were no significant changes in insulin sensitivity or IMCL levels after the low-fat diet. The effects of the high-fat diet showed greater interindividual variation than those of the infusion protocol. The data from the lipid infusion protocol suggest a functional relationship between IMCL levels and insulin sensitivity. Similar effects could be induced by a high-fat diet, thereby underlining the physiological relevance of these observations.